

# Bayesian Engine

# Coming up...

- Estimating the prevalence of Aphantasia
- Manually computing probabilities
- Writing a simple Bayesian model

# A note

Bayesian models might seem scary at the beginning, with all those distributions and math symbols.

But, thankfully we don't have to solve almost any of the math in the background. We just need to **learn how to express theory into models**.

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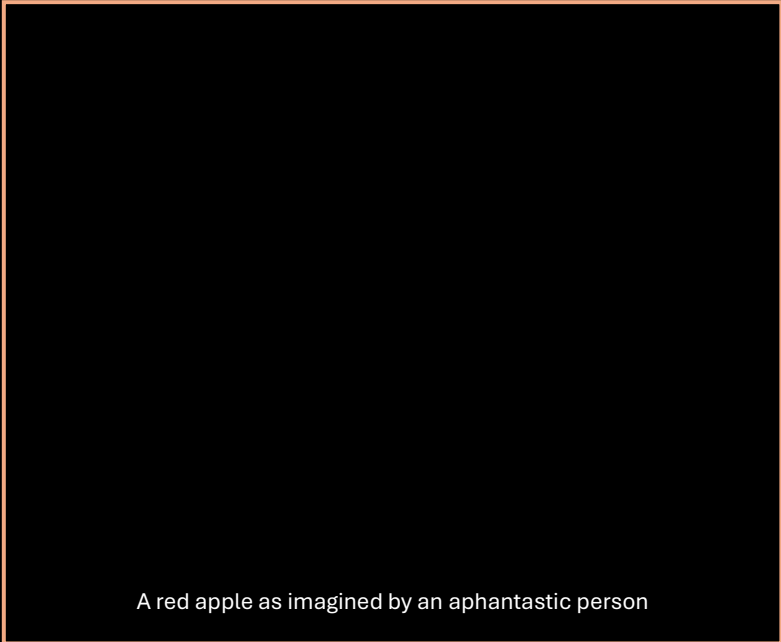
Last year was bad  
I let myself get  
Sucked in by arithmetic



# Aphantasia

Inability to bring images to the “mind’s eye” (Zeman et al., 2015)

- Proportion of people with aphantasia is relatively low:
- 4.2% undergrad students (Dance et al., 2022)
- 0.7% general population (Zeman et al., 2020)
- 4.6% - 14.1% general population (Faw, 2009)



A red apple as imagined by an aphantastic person

# Aphantasia

- Our goal is to estimate the proportion of people with aphantasia.  
How can we do this?

- 1.

- 2.

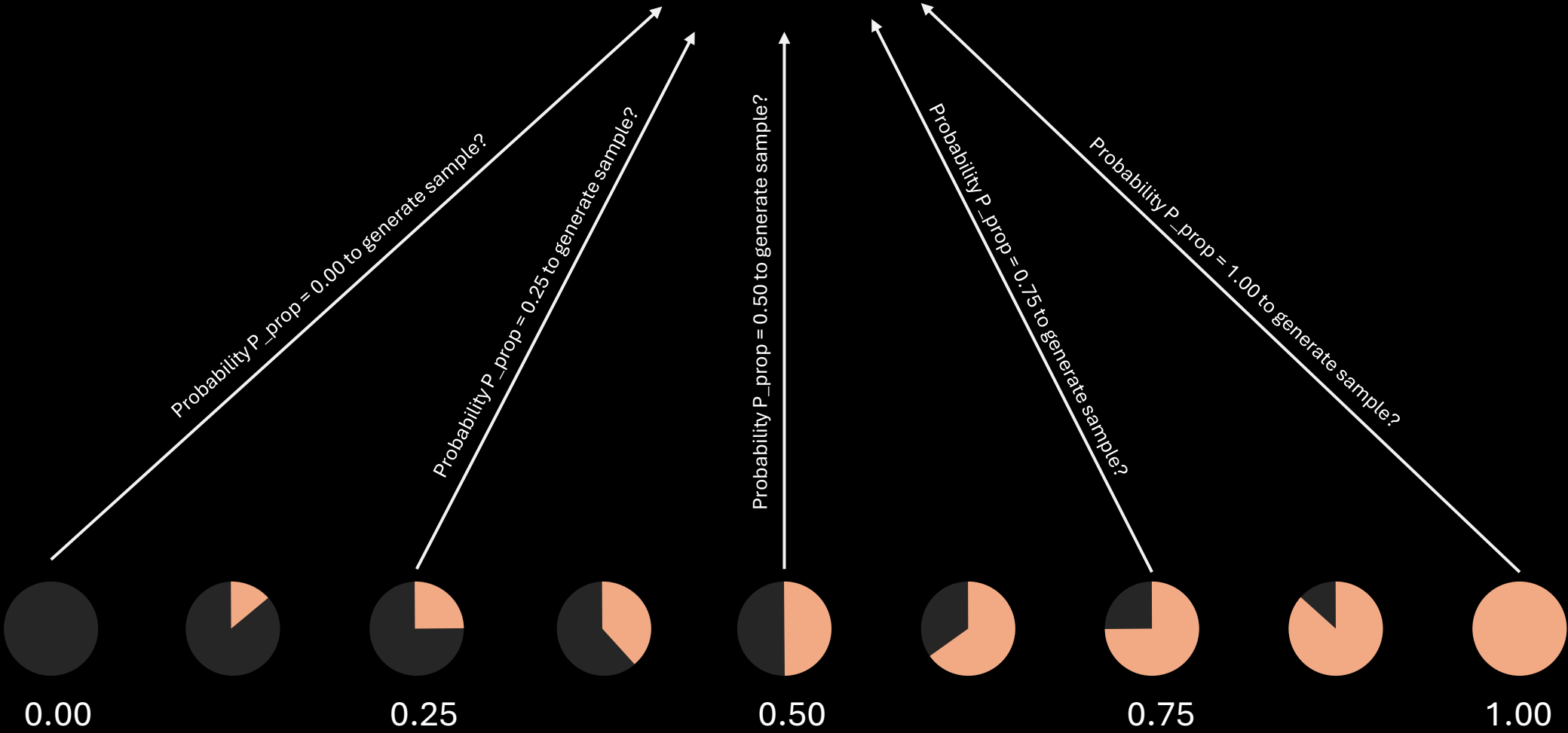
# Aphantasia

- Our goal is to estimate the proportion of people with aphantasia. How can we do this?
1. Survey the population (aka sample)
  2. For every possible true proportion of aphantasia in the population, find how likely the observed data is

We'll start simple, considering only some possible true proportions before generalizing to the anything from 0 (no one) to 1 (everyone).

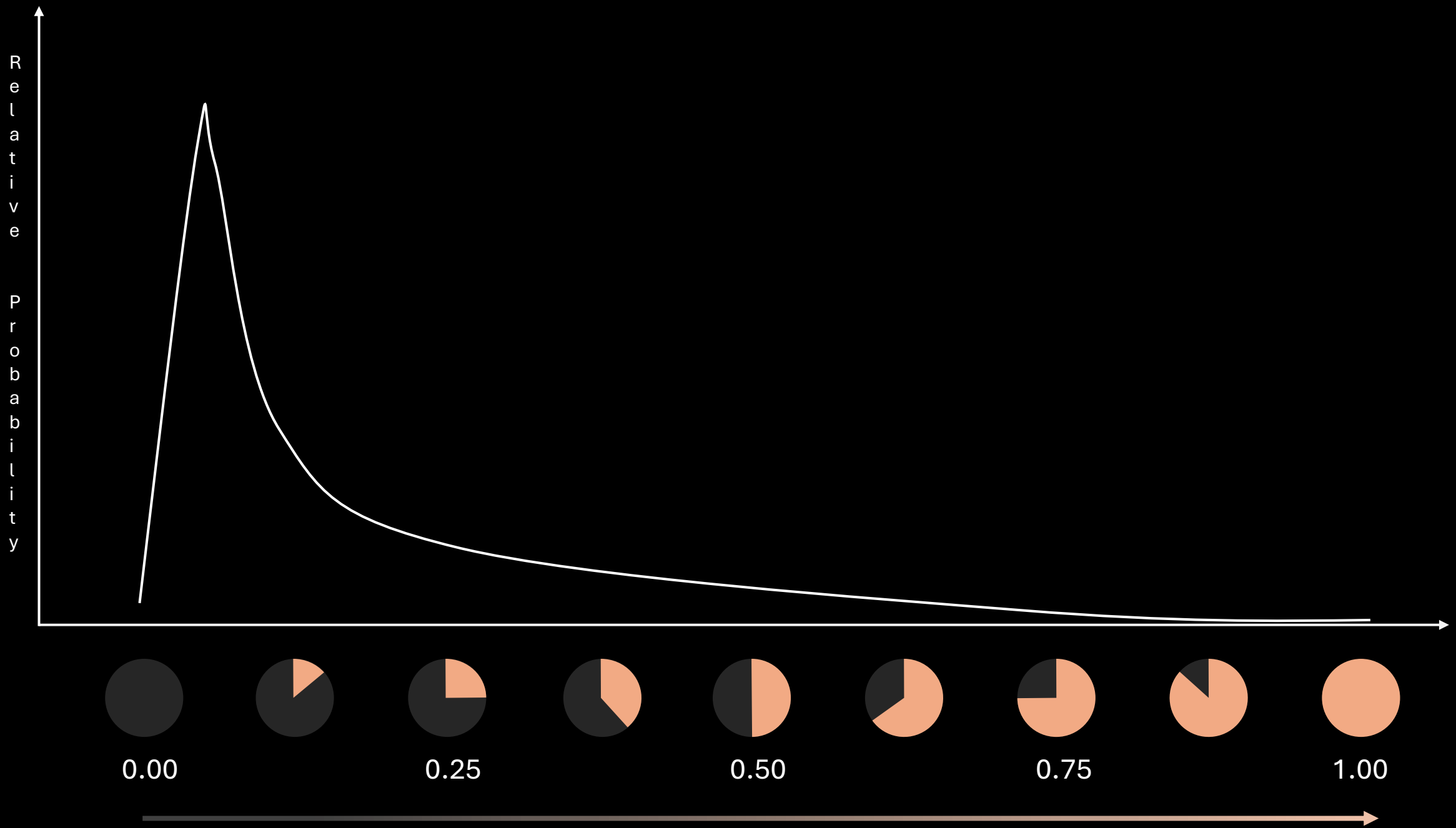
The example is an adaptation of the Globe Tossing Model by Richard McElreath

Survey



Possible proportion of aphantasia in the population [P\_prop]

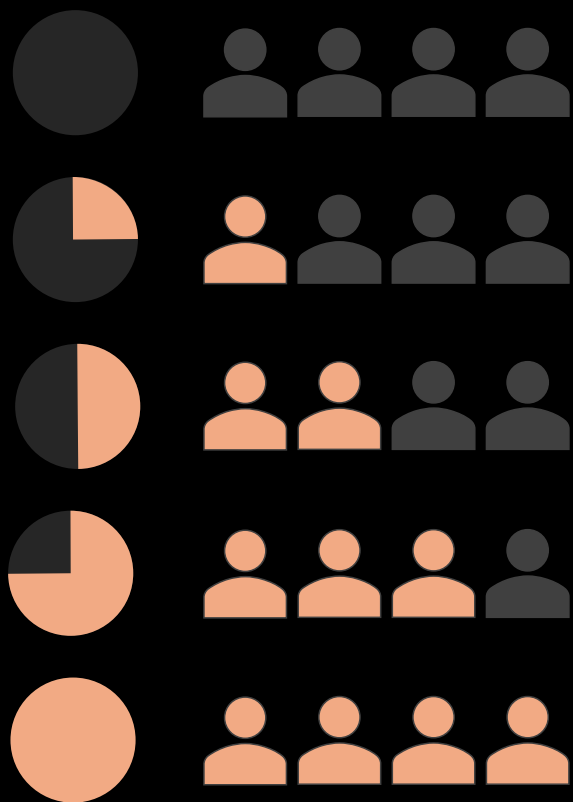




# The long inefficient way: counting



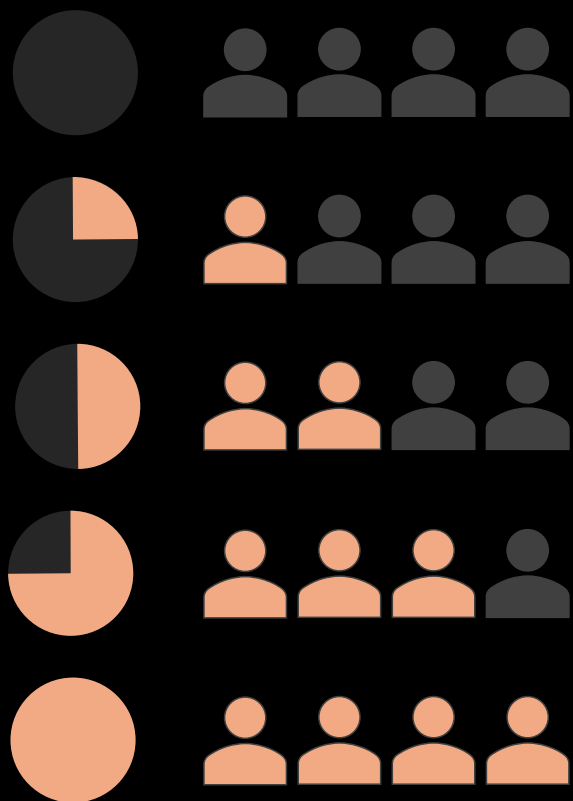
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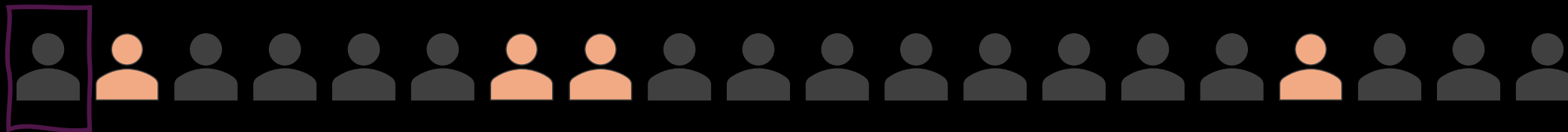




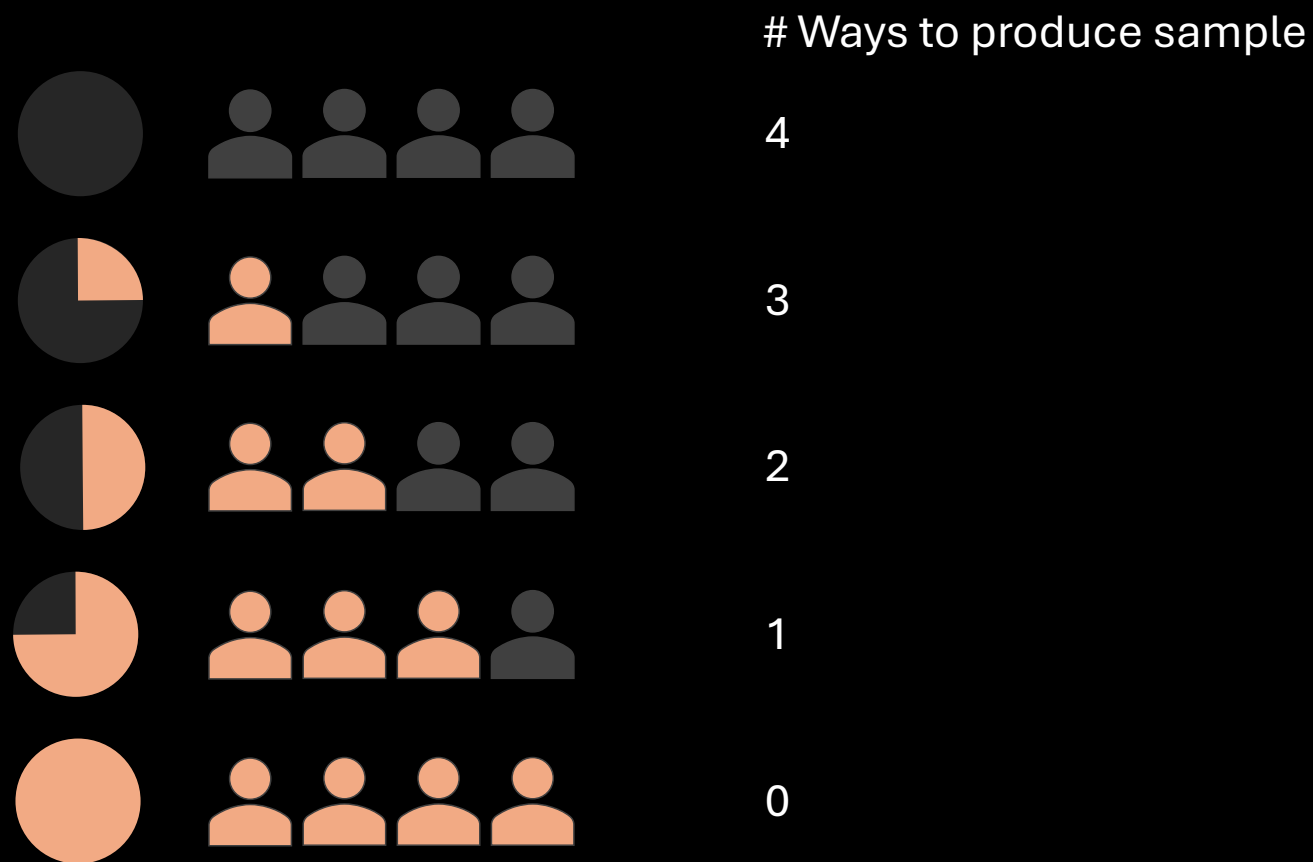
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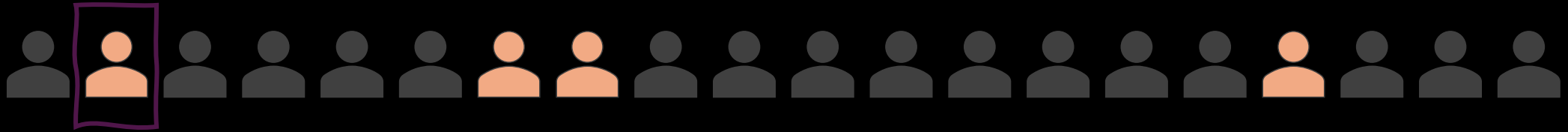
# Ways to produce sample





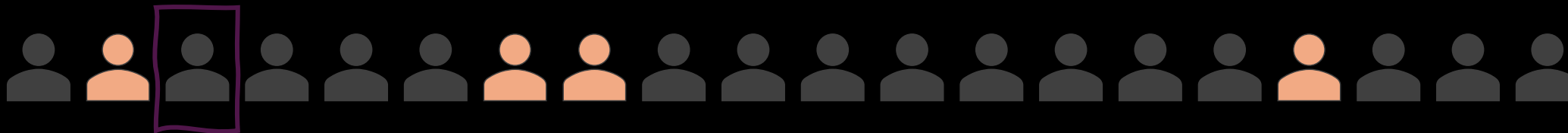
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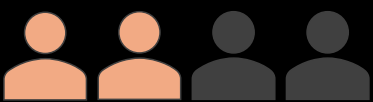
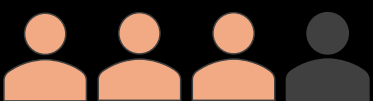
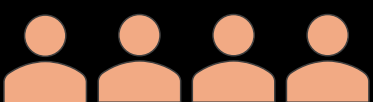


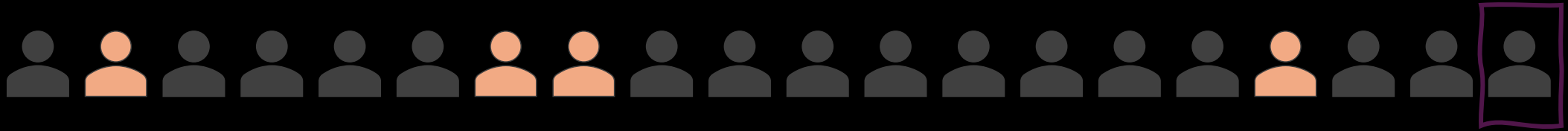
# The long inefficient way: counting

|  |  | # Ways to produce sample |   |
|--|--|--------------------------|---|
|  |  | 4                        | 0 |
|  |  | 3                        | 1 |
|  |  | 2                        | 2 |
|  |  | 1                        | 3 |
|  |  | 0                        | 4 |



# The long inefficient way: counting

|                                                                                     |                                                                                     | # Ways to produce sample |   |   |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------|---|---|
|    |    | 4                        | 0 | 4 |
|    |    | 3                        | 1 | 3 |
|    |    | 2                        | 2 | 2 |
|   |   | 1                        | 3 | 1 |
|  |  | 0                        | 4 | 0 |



# The long inefficient way: counting

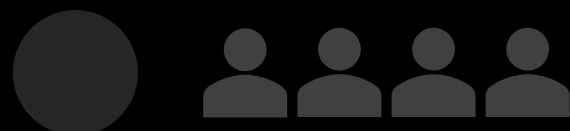
|  |  | # Ways to produce sample |   |   |     |   |
|--|--|--------------------------|---|---|-----|---|
|  |  | 4                        | 0 | 4 | ... | 4 |
|  |  | 3                        | 1 | 3 | ... | 3 |
|  |  | 2                        | 2 | 2 | ... | 2 |
|  |  | 1                        | 3 | 1 | ... | 1 |
|  |  | 0                        | 4 | 0 | ... | 0 |





# The long inefficient way: counting

# Ways to produce sample



$$4 \times 0 \times 4 \times \dots \times 4 = 0$$



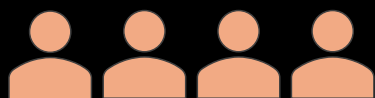
$$3 \times 1 \times 3 \times \dots \times 3 = 43046721$$



$$2 \times 2 \times 2 \times \dots \times 2 = 1048576$$



$$1 \times 3 \times 1 \times \dots \times 1 = 81$$



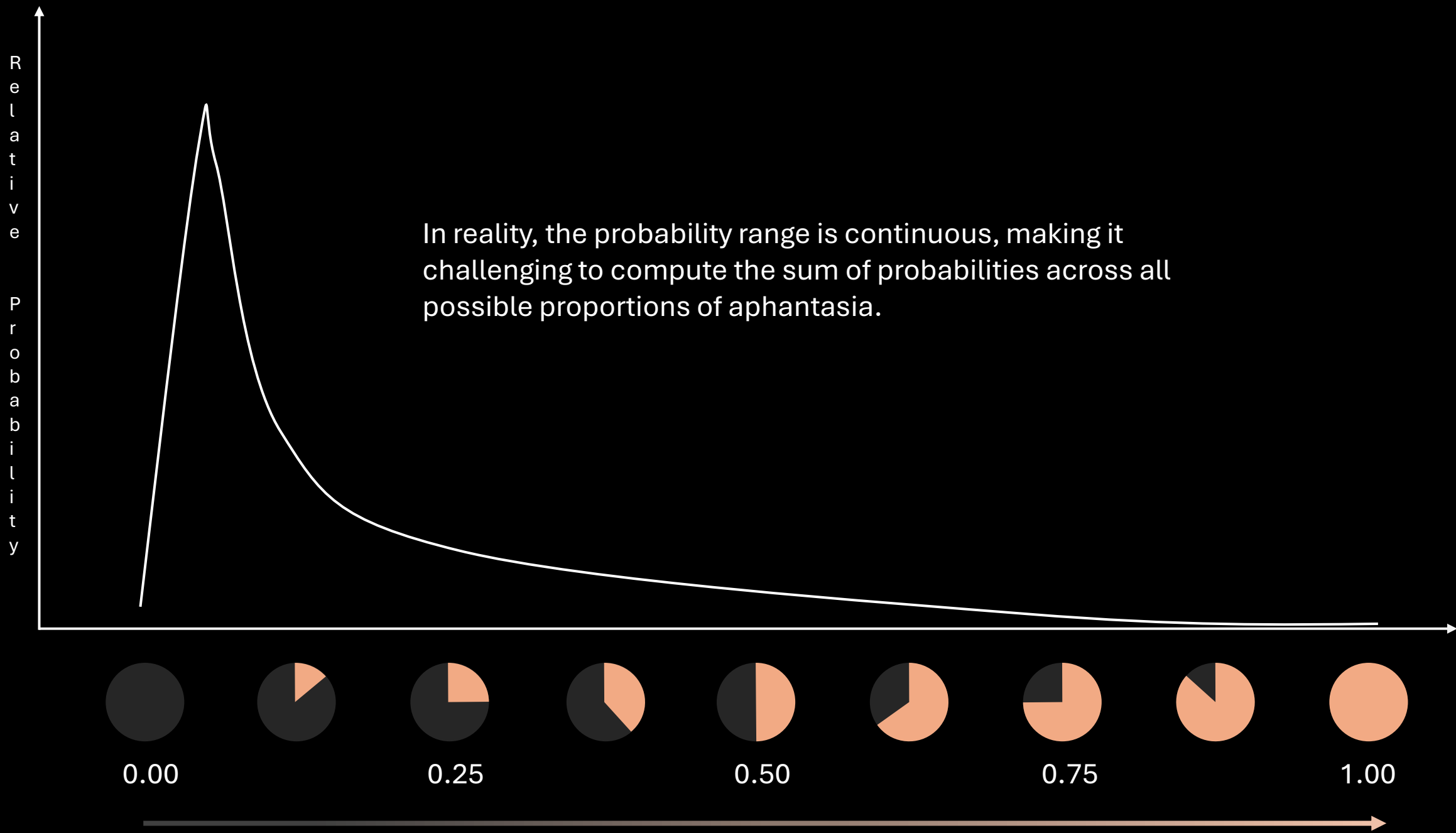
$$0 \times 4 \times 0 \times \dots \times 0 = 0$$

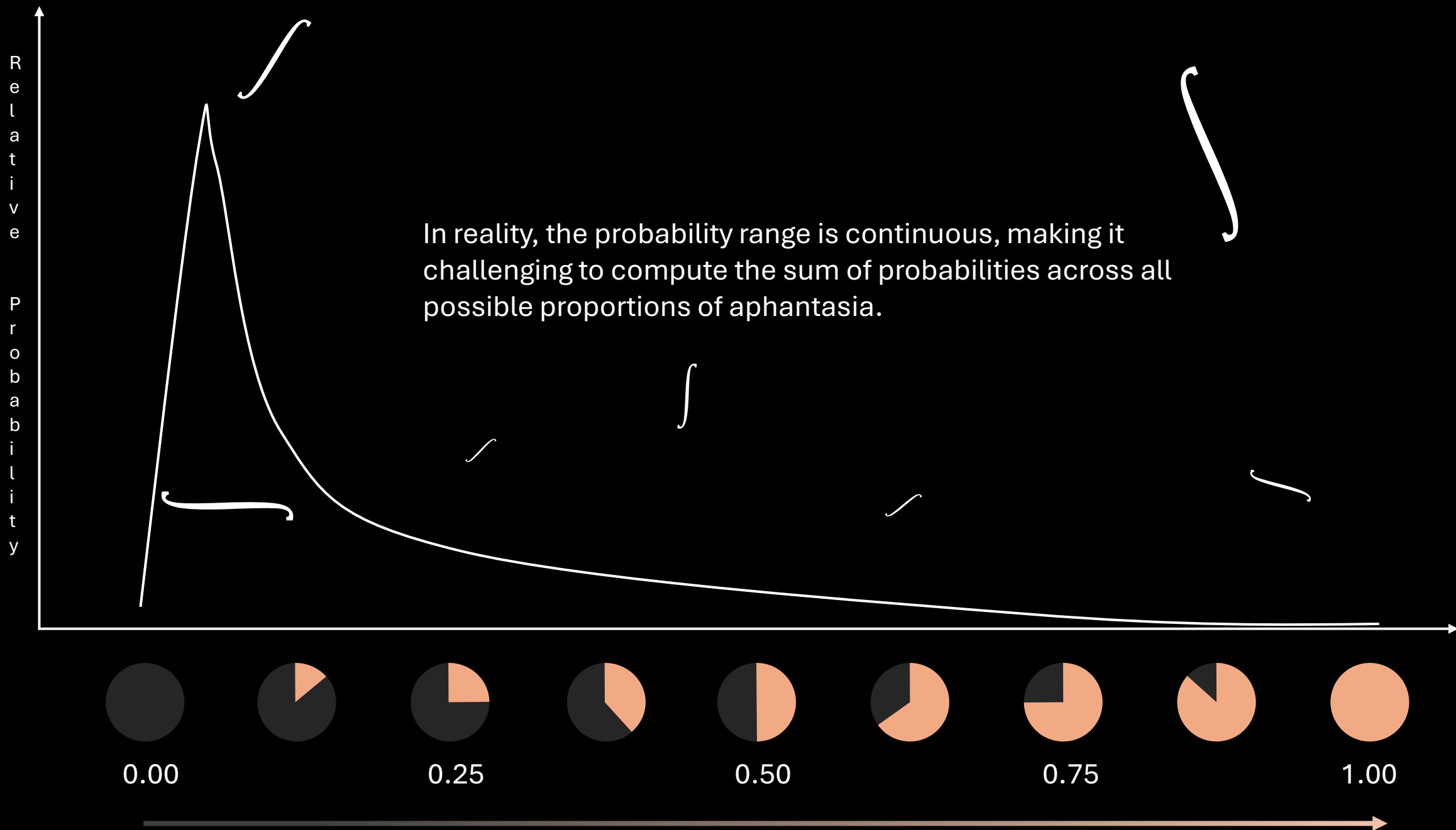


# The long inefficient way: counting

|                                                                                                                                                                         | # Ways to produce sample | # Probability                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------|
|       | 0                        | $0 / 44095378 = 0$                  |
|       | 43046721                 | $43046721 / 44095378 \approx 0.976$ |
|       | 1048576                  | $1048576 / 44095378 \approx 0.023$  |
|     | 81                       | $81 / 44095378 \approx 0.001$       |
|   | 0                        | $0 / 44095378 = 0$                  |

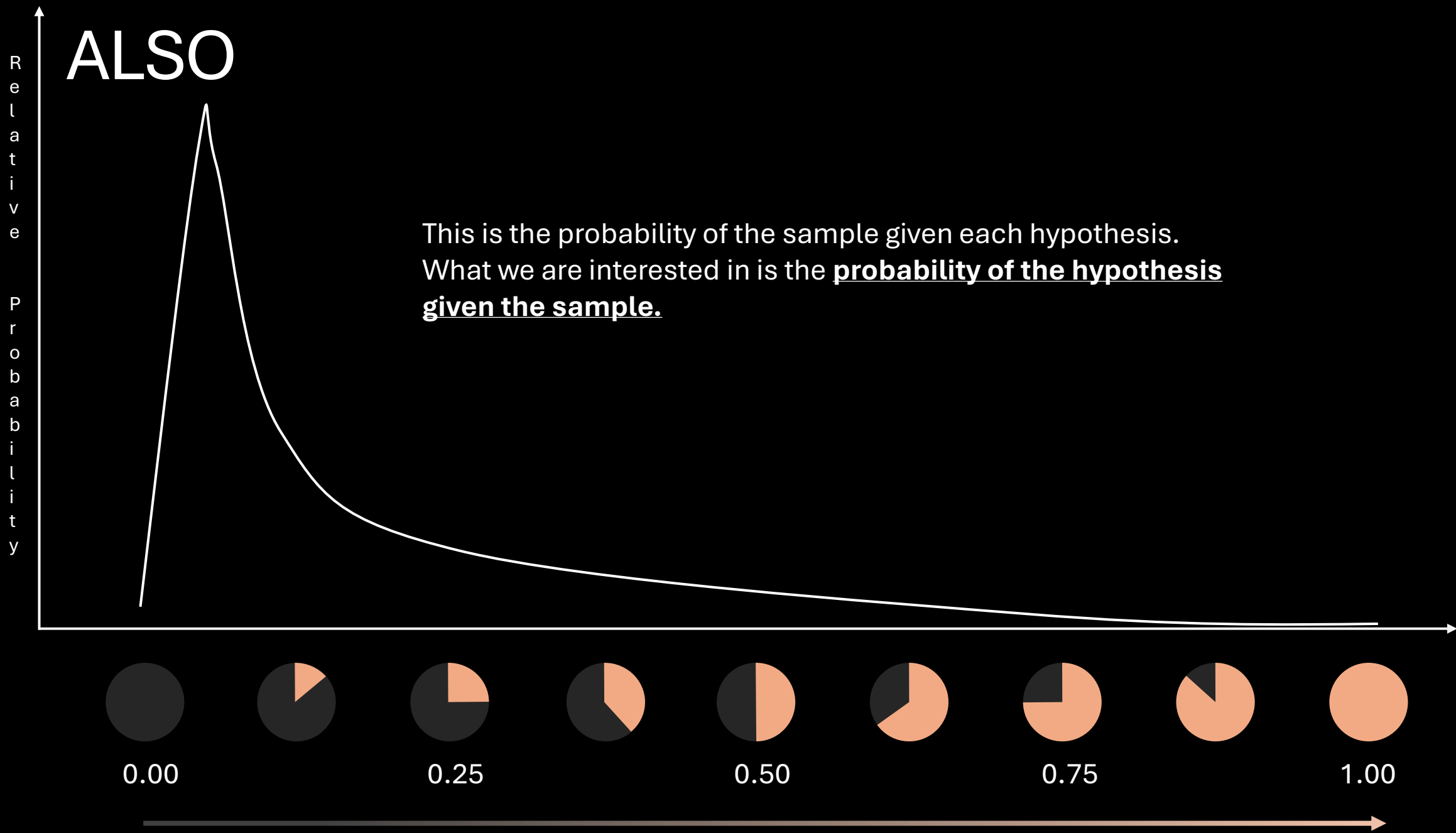
TOT = 44095378





SOLUTION = SAMPLING\*

\*I'm simplifying here



# Bayes Theorem

$$P(A_{population}|A_{sample}) = \frac{P(A_{sample}|A_{population}) \times P(A_{population})}{P(A_{sample})}$$

# Bayes Theorem

$$P(A_{population} | A_{sample}) = \frac{\overset{\text{Likelihood (what we have computed so far)}}{P(A_{sample} | A_{population})} \times \overset{\text{Prior (we have to define it)}}{P(A_{population})}}{\underset{\substack{\text{Average Probability of the Data} \\ \text{Marginal Likelihood}}}{P(A_{sample})}}$$



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$$P(A_{population} | A_{sample}) = \frac{\overset{\text{Likelihood (what we have computed so far)}}{P(A_{sample} | A_{population})} \times \overset{\text{Prior (we have to define it)}}{P(A_{population})}}{\underset{\text{Average Probability of the Data}}{\underset{\text{Marginal Likelihood}}{P(A_{sample})}}}$$

Averaged over the prior  
Formally, it is the integral of the probability of Aphantasia in the sample over all the possible proportions of aphantasia in the population

LET'S CODE THIS

